

Lesson 03: Separable First-Order Differential Equations

Professor Dave Explains Differential Equations Master Series · Lesson 3

Video: Separable First-Order Differential Equations **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave highlights separable equations as the cleanest analytical tool in differential equations. If all dependent terms y can be quarantined on the left with dy , and all independent terms x placed on the right with dx , the problem reduces to two independent single-variable calculus integrals. However, Dave warns of the classic exam trap: whenever you divide by an expression involving y , you risk throwing away constant equilibrium solutions where that divisor equals zero!

2. Core Theoretical Framework & Rigorous Formulations

Analytical Separation Mechanism

A first-order equation is separable if it factors as:

$$\frac{dy}{dx} = g(x)h(y) \implies \frac{1}{h(y)} dy = g(x) dx \quad (h(y) \neq 0)$$

- **Singular Solutions:** Any real constant $y = c$ such that $h(c) = 0$ is an equilibrium solution that satisfies the ODE identically.
- **Explicit Branch Selection:** When taking square roots or logarithms, the initial condition $y(x_0) = y_0$ dictates the proper sign ($\pm$) and branch.

3. Fully Solved Whiteboard Problems

Exam Problem: Solving a Nonlinear Separable IVP with Branch Selection

Problem Statement: Solve the initial value problem $\frac{dy}{dx} = \frac{2x+1}{2y}$, with $y(0) = -3$. State the explicit solution and its interval of definition.

Step 1: Separate Differentials

$$2y \, dy = (2x + 1) \, dx.$$

Step 2: Direct Indefinite Integration

$$\int 2y \, dy = \int (2x + 1) \, dx \implies y^2 = x^2 + x + C.$$

Step 3: Enforce Initial Condition

$$y(0) = -3 \implies (-3)^2 = 0 + 0 + C \implies C = 9.$$

Step 4: Solve for $y(x)$ and Select Root Branch

$$y(x) = \pm \sqrt{x^2 + x + 9}. \text{ Since } y(0) = -3 < 0, \text{ we MUST select the negative square root branch: } y(x) = -\sqrt{x^2 + x + 9}.$$

Step 5: Determine Domain of Existence

The radicand $x^2 + x + 9$ has discriminant $\Delta = 1 - 36 = -35 < 0$, so it is strictly positive for all $x \in \mathbb{R}$. Interval of existence: $(-\infty, \infty)$.

⚠ Common Exam Pitfall & Trapdoor Avoidance

Leaving $\pm$ in your final answer on a Concordia quiz or midterm will lose marks. An initial condition picks exactly one branch of the square root.

4. Key Takeaways & Exam Strategy Formula Card

🔑 Core Takeaway & Exam Rule

Separate, integrate, add $+C$ immediately, and let the initial value dictate the explicit root sign.

- Video reference: [Separable First-Order Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.